



Manzil One

CRM SUITE

Product Documentation & PRD

End-to-end: how it works, the quotation engine, the AI, and what is / isn't live

LIVE APPLICATION

manzilone.vercel.app

SOURCE REPOSITORY

github.com/SyedAbdulKareem13/manzilone

STACK

Next.js 15 - Prisma - Supabase - NextAuth - Gemini -
Vercel

UPDATED

June 2026 (v2)

A premium CRM & quotation platform for consulting, staffing, manpower, licensing and project businesses.

1. Executive summary

Manzil One is a production-deployed multi-tenant SaaS CRM that runs the full revenue lifecycle in one workspace: Leads -> Opportunities -> RFQ -> Quotation -> Approval -> Close. On top of the core pipeline it adds a manpower / non-manpower / license rate-card master, a Position Determination Engine for resource estimation, a multi-step quotation approval workflow with a field-level audit trail, an activity timeline on every lead and opportunity, a no-code field-configuration studio in Admin, revenue analytics, branded PDF export, and **Manz AI** - a built-in assistant that can draft an entire rate-card-priced quotation from a one-line brief.

This v2 of the documentation reflects the current production build. Since v1 the product was renamed and re-homed (now live at manzilone.vercel.app, canonical repo 'manzilone'), Activities moved from partial to fully working (timeline + log + edit on leads and opportunities), a no-code Configuration studio was added, quotations became editable with a transparent audit trail, and the Manz AI workspace and AI quote drafting shipped.

2. Live environment & access

Live URL	https://manzilone.vercel.app
Repository	github.com/SyedAbdulKareem13/manzilone (branch: main). The older 'crm-mini' repo is archived (read-only).
Hosting	Vercel - serverless functions pinned to Tokyo (hnd1), colocated with the database. Deployed via CLI from the local folder.
Database	Supabase PostgreSQL (ap-northeast-1, Tokyo) via Prisma + pgBouncer pooler. One shared DB.
AI	Google Gemini via a REST client; key + model stored in DB (AppConfig) and swappable without redeploy.

Seeded demo accounts (password for all: password123):

Email	Role	Can do
admin@nova.crm	ADMIN	Everything incl. Admin master data, Configuration & approvals
manager@nova.crm	SALES_MANAGER	Pipeline, approvals (step 2)
sales@nova.crm	SALES_EXEC	Leads, opportunities, RFQ, quotations, activities
finance@nova.crm	FINANCE	Final approval step

Note: login emails remain on the @nova.crm seed domain (the product was renamed after seeding; the data layer was not re-keyed to avoid disrupting live logins).

3. Architecture & how a request flows

A single Next.js 15 App-Router project. Pages under /app are server components that read the session and query PostgreSQL through Prisma; interactive pieces (dialogs, kanban, pickers, the activity panel, the AI workspace) are client components. Mutations go through REST route handlers under /api that re-check auth and validate with Zod.

- **Middleware** guards /app/*: checks the Auth.js session cookie (incl. chunked variants), redirects to /login if absent.
- **Server render**: the matched server component calls auth() (role & organizationId ride inside the signed JWT - no per-request DB hit), then runs org-scoped Prisma queries.
- **Data**: Prisma talks to Supabase over the IPv4 pooler (6543, pgbouncer) at runtime and the session pooler (5432) for migrations; compute + DB both in Tokyo (~1-2ms/query).
- **Mutations**: client components POST/PATCH to /api/* handlers; the UI updates optimistically and calls router.refresh(). Sensitive writes record an audit-log entry.
- **Config-driven forms**: form fields read a per-org FieldConfig table (label / shown / required / order) so Admin can reshape forms with no redeploy.

4. Technology stack

Layer	Technology	Purpose
Framework	Next.js 15.5 (App Router, RSC)	Routing, SSR, route handlers
Language	TypeScript (strict)	End-to-end type safety
UI	Tailwind CSS + shadcn/Radix primitives	Design system, accessibility
Motion	Framer Motion	Kanban drag, transitions, the Manz orb
Charts	Recharts (lazy-loaded)	Dashboard analytics
Auth	NextAuth v5 (Auth.js) + bcrypt	Google / password / OTP, sessions
ORM / DB	Prisma 5 + Supabase PostgreSQL	Data model, queries, migrations
AI	Google Gemini (REST client, lib/ai.ts)	Chat, RFQ parse, scope text, drafting
Hosting	Vercel + GitHub	Build, deploy, preview

5. Core data model (key additions in v2)

All business records carry an organizationId for multi-tenancy. Alongside the core entities (Organization, User, Territory, BusinessUnit, Customer, Contact, Lead, Opportunity, RFQ, Quotation, PositionEstimate, the three Rate Cards, ApprovalChain / Request / Step), v2 adds:

- **FieldConfig** - per-org, per-module field settings (label, active, required, position, custom fields). Drives form rendering.
- **ActivityTypeConfig** - configurable activity types (label, icon, colour, active, order).
- **Activity.data (JSON)** - values for admin-defined custom activity fields.
- **AuditLog** - field-level change trail (entity, action, actor, summary, before/after) shown on records.
- **AppConfig** - singleton settings incl. the Gemini API key/model and the landing hero version.

6. Quotation - the latest working (in depth)

The quotation is the heart of the product. Today it works end-to-end three ways - built manually, built from an RFQ, or drafted by AI - and every quote is editable with a transparent audit trail and a multi-step approval.

Building & pricing

- Build from an RFQ (carries customer + line items) or from scratch. Line items are typed Manpower / Non-Manpower / Software License / Hardware / Service.
- Per line set unit cost, quantity, markup %, discount % and tax %; base cost, grand total, profit and margin recompute live as you type.
- **Position Determination Engine**: estimates headcount x duration x monthly rate into cost, monthly billing, revenue, margin and margin % per role - the resourcing view behind the price.
- Rate cards feed pricing: manpower designations, license products and non-manpower categories supply the rates.

Draft with AI

- A one-line brief (e.g. '3-month Salesforce CRM for a healthcare provider; 1 PM, 2 Developers, 1 QA; Fixed Price') becomes a complete draft quotation - team, line items, positions, pricing and margin.
- Each role is priced from your manpower rate cards by closest-designation match; unmatched roles use indicative defaults, and the quote's terms state which.
- Drafts are flagged with an 'AI' mark in the Quotations list and land as a fully editable DRAFT.

Editing, audit & approval

- Quotes are editable after creation - change line items, pricing and terms with live totals; a status guard protects approved quotes.
- Every change is captured in a field-level **audit trail** (who changed what, when) visible on the record.
- A four-step approval workflow (Sales Exec -> Sales Manager -> Business Head -> Finance) supports submit / approve / reject with comments.
- Export a branded, print-optimised A4 PDF (Najm-star letterhead, parties, line items, totals, terms).

7. Manz AI - how it works & why it's useful

Manz AI is the built-in assistant (workspace at /app/ai, plus a one-click launcher in the top bar that uses the animated 'Manz orb'). It is designed so that the structure and money in a quote are always deterministic and reliable - the AI only adds language, never the numbers.

What it does

- **Draft a quotation** from a plain-English brief (the headline capability).
- **Assistant chat** - workspace-aware Q&A about the pipeline.
- **RFQ parser** - turns pasted RFQ text into structured line items.
- **Company overview** - an AI brief + talking points for a prospect.
- **Email writer** - drafts follow-up / proposal emails.

How the quote drafting works (the important part)

1. **Parse (rules, no AI)**: a deterministic parser extracts duration, pricing model, scope/title, client and the team (counts x roles). Works with no AI key.
2. **Match to rate cards**: each role is scored against your manpower rate-card designations - exact, substring, then token overlap requiring a distinguishing (non-generic) word, so 'AI Engineer' will not match 'QA Engineer' on 'engineer' alone. A match uses your card's monthly rate; otherwise an indicative default.
3. **Position engine**: billing = rate x 1.35 (35% markup); cost = rate x headcount x months; revenue, margin and margin % follow; lines carry 18% tax. Totals are computed by the same engine the manual builder uses.
4. **Gemini (optional, prose only)**: writes a 2-3 sentence scope paragraph; if the key is missing or rate-limited it is skipped silently. It never touches the figures.
5. **Persist**: find-or-create the customer, create an editable DRAFT quotation (items + positions), note whether rates were matched or indicative, and record an audit entry.

Why it's useful

- **Speed** - a complete, costed, margin-aware draft in seconds instead of building line-by-line.
- **Grounded pricing** - numbers come from your own rate cards, not a guess; the quote is honest about any indicative fallback.
- **Consistency** - the same Position Engine and markup/tax rules apply whether a human or the AI starts the quote.
- **Resilient** - drafting and pricing work even with no AI key; the LLM only enriches wording.
- **Editable + governed** - AI output is a starting draft that flows through the normal edit, audit and approval path - never an unchecked send.

To switch on the live LLM (chat / RFQ parse / scope text), set a valid Google AI Studio key (AlzaSy...) in `AppConfig.geminiApiKey` - no redeploy. The current placeholder key authenticates but is rate-limited to zero, so today chat/tools return a quota message while drafting (deterministic) works fully.

8. Other modules - what changed in v2

Activities (now fully working)

- A live activity timeline on every Lead and Opportunity detail page: log Call / Meeting / Email / Follow-up / Task / Note, with a premium date-and-time picker, status and any admin-defined custom fields.
- Entries can be **edited** and marked done; each shows the owner and a relative timestamp and is mirrored into the record's audit trail. New entries appear instantly (optimistic).

Admin -> Configuration (new, no-code)

- Per-module field configuration for Leads, Opportunities, Customers, Quotations, RFQs and Activities - show/hide, relabel, mark required, drag-reorder, and add custom fields.
- Activity-type configuration - relabel, recolour, pick an icon, reorder and toggle each type.
- The Lead form already renders live from this configuration; rollout to the other module forms is in progress (see section 10).

Unchanged-and-working

- Dashboard (KPI cards + charts), Leads, Opportunities & Pipeline (11-stage Kanban), RFQ, Rate Cards, Customers, Reports (6), Admin master data, Settings (profile / theme / security), Notifications bell, and branded PDF export - all as in v1.

9. Status matrix - what works, what doesn't

Verified against the live deployment. Legend: WORKING = end-to-end on prod; PARTIAL = usable but incomplete; NEEDS SETUP = code ready, needs credentials/config.

Area	Capability	Status
Auth	Email+password, signup, sessions, route guard, OTP, forgot/change password	WORKING
Auth	Google sign-in (needs OAuth client) & real OTP/email/SMS delivery (mock today)	NEEDS SETUP
Leads	List, filter, create, edit, convert; form rendered from field config	WORKING
Opportunities	List, create, edit, clickable stage rail, 11-stage Kanban	WORKING
RFQ	Create (from opp / standalone), typed line items	WORKING
Quotation	Builder, live totals, Position Engine, branded PDF export	WORKING
Quotation	Editable quotes + field-level audit trail	WORKING
Quotation	Draft with AI (rate-card-priced from a one-line brief)	WORKING
Quotation	Four-step approval (submit / approve / reject)	WORKING
Activities	Timeline on leads & opps: log + edit + mark done, audit-linked	WORKING
Admin	Users/roles, territories, BUs, approval chain	WORKING
Admin	Configuration studio: per-module field + activity-type config	WORKING
Rate Cards	Manpower / Non-MP / License CRUD	WORKING
Customers / Reports / Notifications / Settings	Masters, 6 reports, bell, profile/theme/security	WORKING
Manz AI	Workspace UI: chat, RFQ parse, company overview, email writer	WORKING (UI)
Manz AI	Live LLM responses (chat / parse / scope text)	NEEDS SETUP
Config	Custom-field VALUE capture beyond Activities; config-driven forms beyond Lead	PARTIAL
Notifications	Real-time push (currently fetch-on-open)	PARTIAL

Search	Global / AI search (palette is nav + quick-create)	PARTIAL
Documents	File upload / preview / versioning (model + buttons only)	PARTIAL
Profile	Avatar storage (base64 in DB; move to object storage)	PARTIAL

10. What still needs to be developed

- **Turn on the live LLM:** add a quota-enabled Google AI Studio key (AlzaSy...) in AppConfig.geminiApiKey to enable chat, RFQ parsing and AI scope text. Drafting already works without it.
- **Config-driven forms everywhere:** extend the live field-config rendering (already on the Lead form) to the Opportunity, Customer, Quotation and RFQ forms.
- **Custom-field values across modules:** persist and display admin-defined custom fields on every record (today value capture is complete for Activities).
- **AI activity creation & reminders:** log the right activity/task from a sentence, plus due-date nudges (roadmap).
- **Google OAuth + real OTP/email/SMS delivery:** set credentials / wire a provider (logic is production-ready).
- **Document management:** add object storage (Supabase Storage / S3 signed uploads) + a version row per upload.
- **Global full-text / AI search:** a search API (Postgres FTS or pgvector) across leads/opps/RFQs/quotations/customers.
- **Real-time notifications:** SSE / websockets / Supabase Realtime for push (today created on events, fetched on open).
- **Avatars to object storage;** audit logging + rate-limiting on auth endpoints.
- **Automated tests:** Playwright e2e + unit tests on the pricing / Position Engine and the AI rate-card matcher.

11. Run, seed & deploy

Local (Node 20+):

```
npm install
cp .env.example .env # set DATABASE_URL, DIRECT_URL, AUTH_SECRET
npm run db:push && npm run db:seed
npm run dev
```

- **Seeding:** 'npm run db:seed' (needs DATABASE_URL set) OR paste prisma/seed-supabase.sql into the Supabase SQL Editor (no local tooling). Both replace the org's leads/opps/RFQs/quotes/activities with demo-ready, current-dated data; safe to re-run.
- **Deploy:** 'vercel deploy --prod' from the local folder (Tokyo region + build are pre-configured); run db:push once when the schema changes.
- **Ops note:** Supabase DATABASE_URL/DIRECT_URL are marked Sensitive in Vercel, so 'vercel env pull' returns them empty - grab the connection string from Supabase to run migrations/seed locally.

12. Recommended next steps

- Enable the live Gemini key and a real OTP/email provider to clear the two 'needs setup' items.
- Finish config-driven forms + custom-field values across all modules (the differentiator already proven on Leads & Activities).
- Ship AI activity creation and reminders; add document storage and global search.
- Add real-time notification push and automated tests on the pricing / AI engines.

Manzil One - CRM Suite. This documentation (v2) reflects the live build at time of generation and is intended for internal product review.